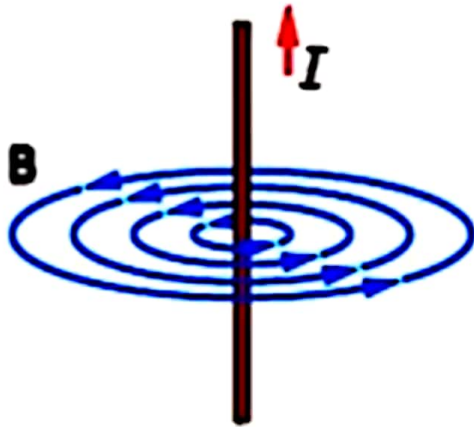




Magnetic Field  Electric Current



YES

Faraday and Henry experimentally proved that, when a coil is placed in variable magnetic field then a current is induced in the coil.

Electromagnetic Induction (EMI)

The phenomena of generating current and e.m.f. by changing magnetic fields is called Electromagnetic Induction (EMI).

Induced e.m.f. & Induced Current

The e.m.f. so developed is called induced e.m.f.

If the conductor is in the form of a closed circuit then a current flows in the circuit. This is called induced current.

* The phenomena of EMI is the basis of working of power generators, dynamos, transformers etc.

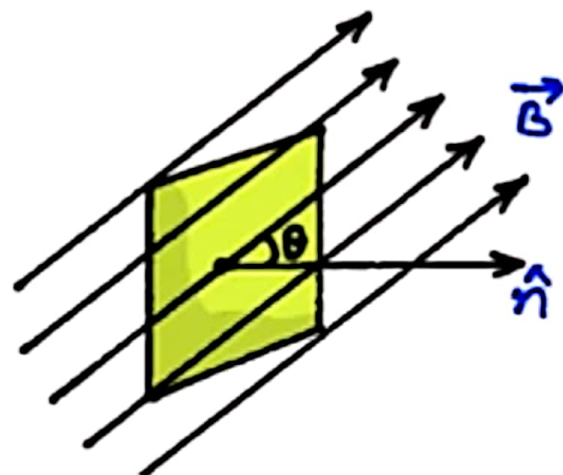
Magnetic Flux (ϕ)

The magnetic flux (ϕ) through any surface held in a magnetic field $\vec{B}$ is measured by the total number of magnetic lines of force crossing the surface normally.

$$\phi = \vec{B} \cdot \vec{A} = BA \cos \theta$$

If the coil has N turns then total amount of magnetic flux linked with the coil is

$$\phi = N (\vec{B} \cdot \vec{A}) = NBA \cos \theta$$



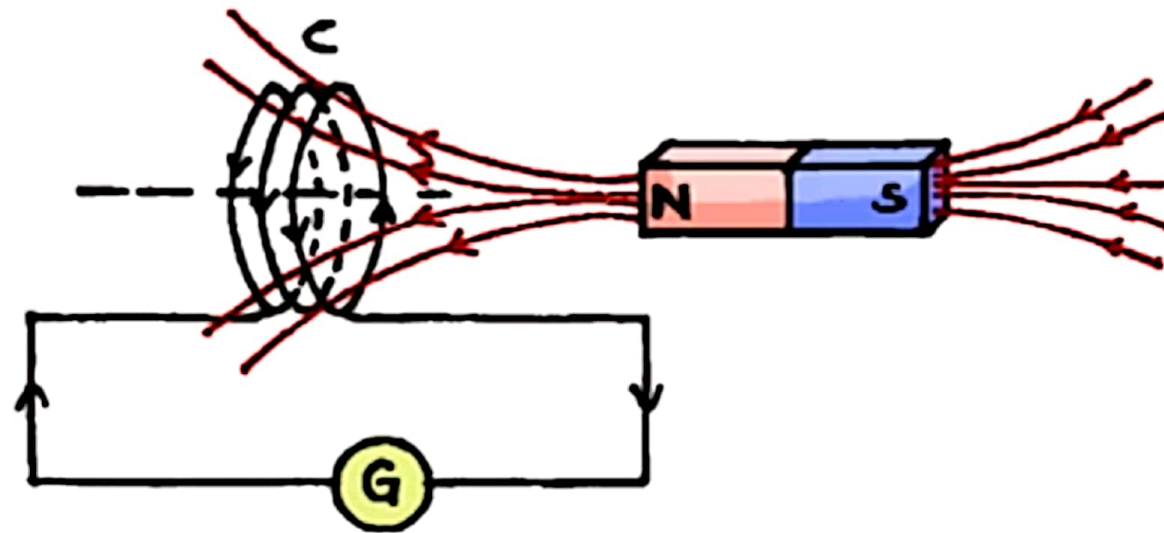
* When $\theta = 90^\circ$ then $\phi = 0$

* When $\theta = 0^\circ$ then $\phi = BA$ (Max value)

SI Unit - Weber (Wb) or $T \cdot m^2$

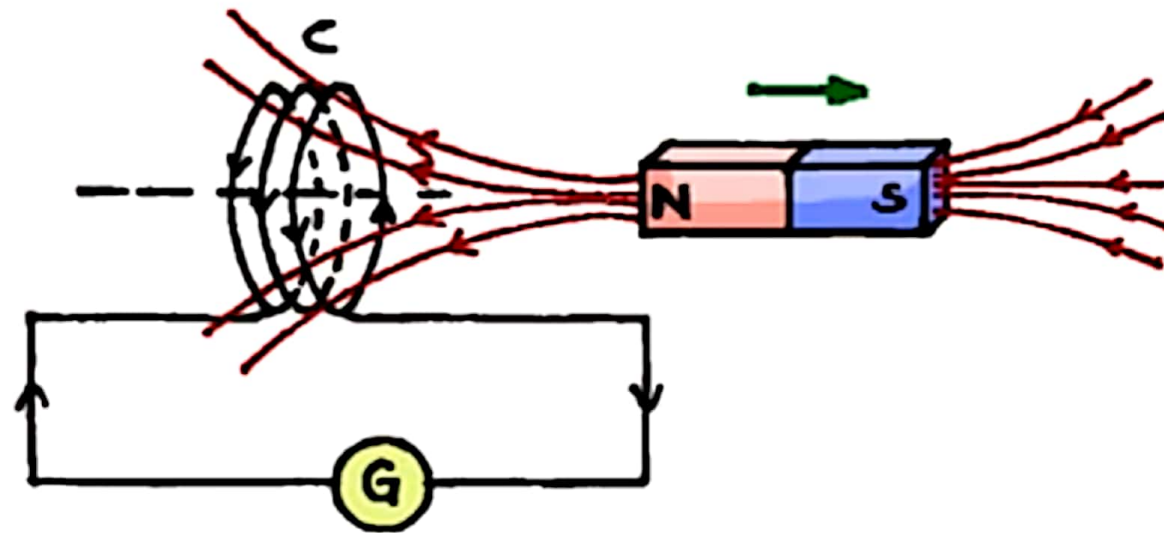
- * Magnetic flux (Φ) is a scalar quantity.
- * When a body is present in a uniform or non-uniform magnetic field, outward flux is taken to be positive, while inward flux is taken as negative.

Experiment - 1 (Current induced by a magnet)



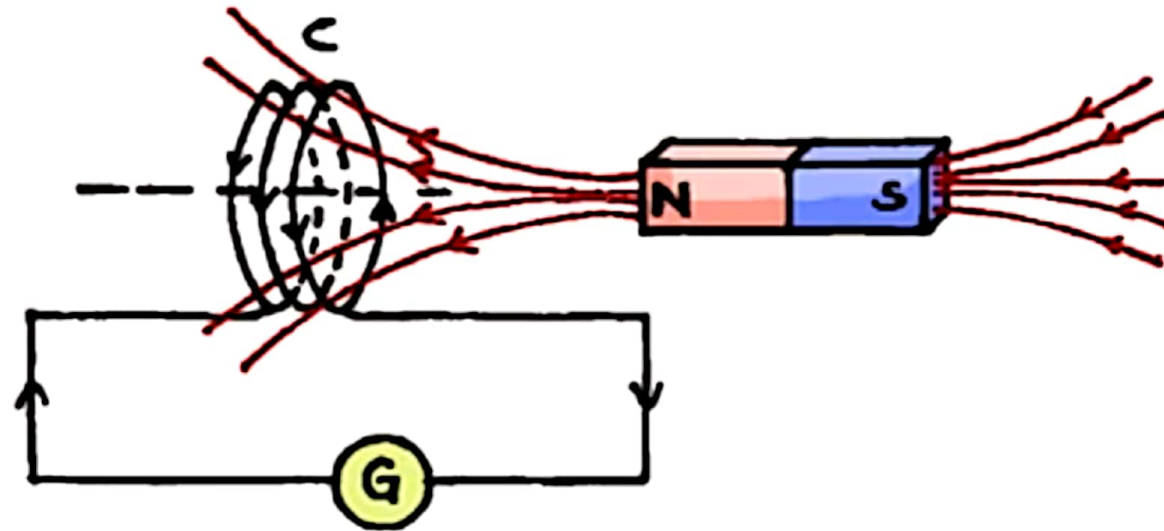
- (i) When north pole of a bar magnet is pushed towards the coil, the galvanometer shows a sudden deflection, which indicates that current is induced in the coil.
- (ii) There is no deflection when the bar magnet is held stationary anywhere.

Experiment - 1 (Current induced by a magnet)



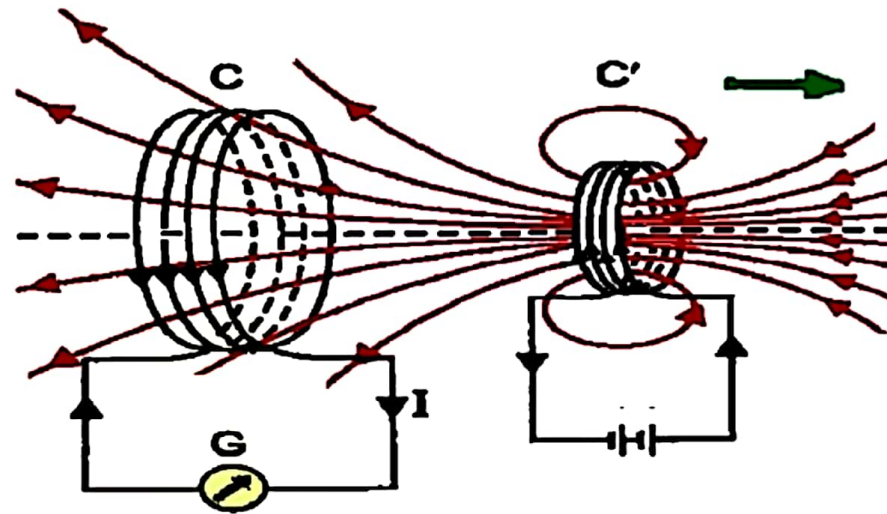
(iii) When the magnet is moved away from the coil, the galvanometer shows deflection in the opposite direction, which indicates reversal in the direction of induced current.

Experiment - 1 (Current induced by a magnet)



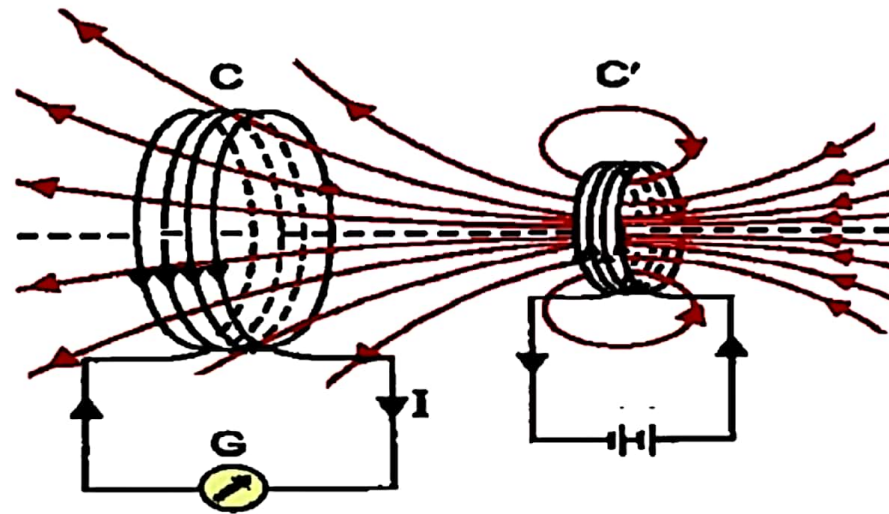
It shows relative motion between the coil and the magnet is responsible for induction of electric current in the coil.

Experiment 2 (Current induced by current)



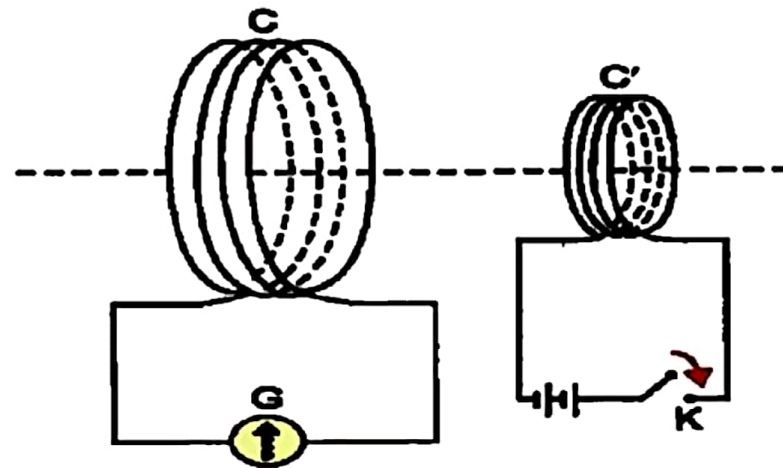
- (i) When coil C' is moved towards the coil C , the galvanometer shows a sudden deflection. This indicates that electric current is induced in the coil C .
- (ii) When the coil C' is moved away from the coil C , the galvanometer shows a deflection in the opposite direction. This indicates that direction of current induced in coil C is reversed.

Experiment 2 (Current induced by current)



- (iii) There is no deflection when there is no relative motion between the two coils.
- (iv) The galvanometer deflection (and hence induced current) is found to be larger when coils are moved faster towards / away from each other.

Experiment 3 - (Current induced by changing Current)

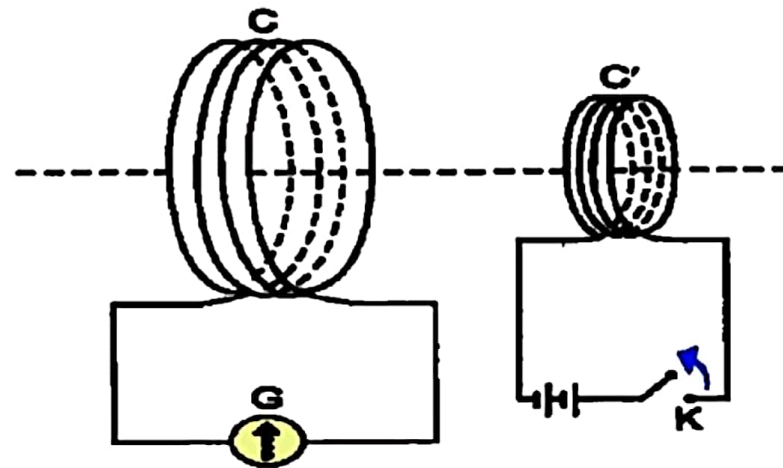


From the experiment 3 Faraday showed that relative motion is not an absolute requirement.

Current can be induced without relative motion.

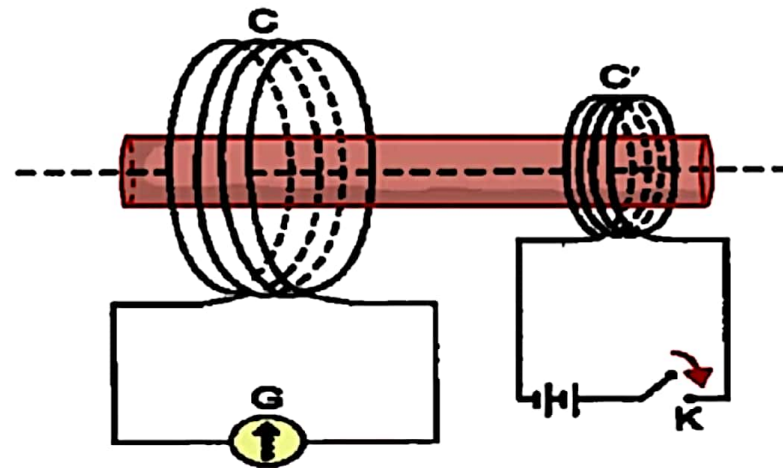
- (i) When Key K is pressed, galvanometer in coil C shows a momentary deflection, indicating that current is induced in coil C.

Experiment 3 - (Current induced by changing Current)



- (ii) When the Key is pressed continuously, there is no deflection in the galvanometer.
- (iii) When the Key is released the galvanometer shows again a momentary deflection but in the opposite direction.

Experiment 3 - (Current induced by changing Current)



(iv) The galvanometer deflection increases, when an **iron rod** is inserted into the coils along their axis, and the Key is pressed / released.

Cause of Induced E.M.F.

All experiment shows that induced e.m.f. appears in a coil whenever the amount of magnetic flux linked with the coil changes with time.

Note- Only the presence of magnetic flux is not enough. The amount of magnetic flux linked with the coil must change in order to produce any induced em.f. in the coil.

More Relative motion
Between Coil and Magnet



More induced EMF

- Direction of induced emf or current depends on the direction of motion of the magnet.